

MTF microdata
(ownership, dayparts)



NASA POWER
GHI, T (LST)



AMDA / SEforALL
cost benchmarks



RURA / EPRA
tariff schedules



Stochastic demand
RAMP appliances +
cooking events (ϕ)



Hourly dispatch
PV $\rightarrow$ load $\rightarrow$ battery
(8,760 h $\times$ 20 yr)



Least-cost sizing
worst-year
unmet $\leq 5\%$



Finance
LCOE, NPV, IRR,
inverse solvers

scenario sweep: penetration $\times \phi \times$ tariff $\times$ discount $\times$ seed